

Name:
Roll Number:
Program: BTech / MTech (2 Year) / MTech (3 Year) / PhD

AI3000: REINFORCEMENT LEARNING

QUIZ 02

Date: 12 September 2026

Question	1	2	Total
Marks Scored			

1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with derivative f' . The Newton–Raphson method to find the root of f is an iterative procedure that starts with some $x_0 \in \mathbb{R}$ and iteratively computes

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

for $k \geq 1$, assuming that $f'(x_k) \neq 0$ for any $k \geq 0$. We can express this procedure as a fixed-point iteration

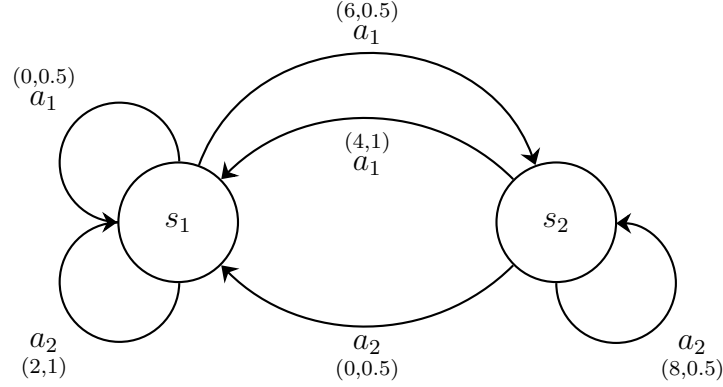
$$x_{k+1} = F(x_k).$$

- (a) **(2 Marks)** Find an f such that F is NOT a contraction mapping.
(b) **(2 Marks)** Find an f such that F is a contraction mapping.

2. Consider an infinite-horizon discounted MDP with state space

$$\mathcal{S} = \{s_1, s_2\}, \quad \mathcal{A} = \{a_1, a_2\},$$

and discount factor $\lambda = 0.5$. The MDP is graphically represented with the state transition diagram below. Circles denote states, arcs denote actions and transitions between states, and the expressions in parentheses denote rewards and transition probabilities, respectively. Zero probability transitions have been omitted.



- (a) (2 Marks) Calculate the expected one-step reward function $r(s, a)$ for every state-action pair.
(b) (2 Marks) Consider the deterministic decision rule $\phi : \mathcal{S} \rightarrow \mathcal{A}$ defined by

$$\phi(s_1) = a_1, \quad \phi(s_2) = a_2.$$

For the above decision rule ϕ , write down P_ϕ and r_ϕ .

- (c) (2 Marks) Calculate $v_{\infty, \lambda}^*$.